

The background is a stylized illustration of a soccer penalty shootout. A goalkeeper in a white jersey and yellow cap is diving to the left in front of a goal. A soccer ball is in mid-air, having just been kicked. A large, dark blue boot with yellow studs is shown in the foreground, as if it has just kicked the ball. The scene is set on a green field with a blue sky and a red goalpost. There are various geometric shapes like stars and circles scattered across the sky.

Web Technology Project

Penalty Shootout Game

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Tools Used:
HTML, CSS, JS, PHP, MySQL

Penalty Shootout Game

- A full-stack web game where you take and save penalty kicks
- Concede a goal? Answer a football trivia question to stay alive
- Built with: HTML, CSS, JavaScript, PHP, MySQL



Problem Statement

- Most browser-based games are client-only — close the tab, everything is gone
- No user accounts, no score history, no way to compare with others
- No way for an admin to manage game content without touching the database



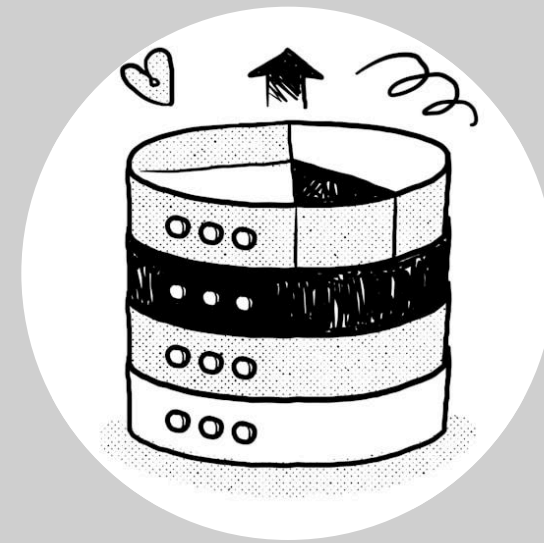
Proposed Solution



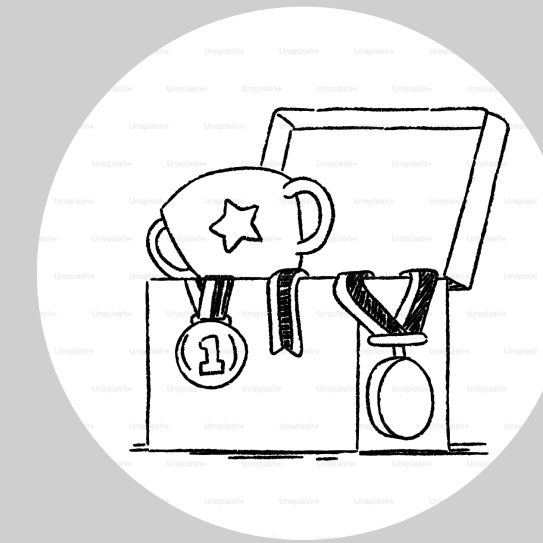
Allow users
to play an
interactive
penalty
game



Implement
user login
and
authentication system



Store and
manage
player
scores in
database



Display
leaderboard
for
performance
tracking

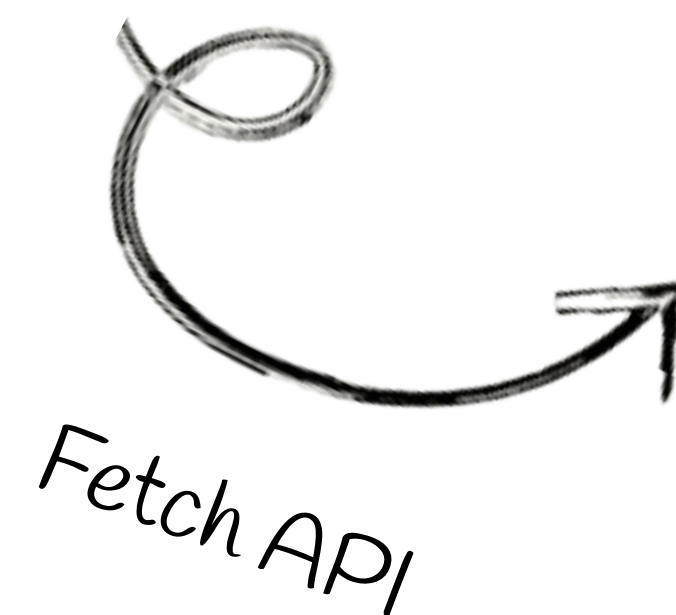


Provide
admin panel
for system
management

System Architecture



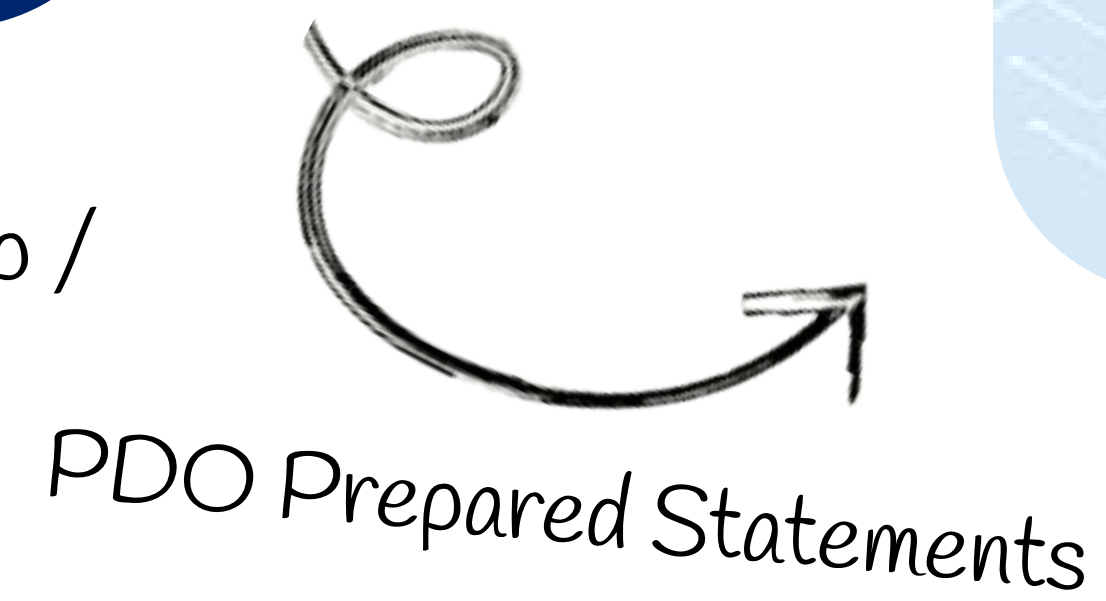
Browser (HTML +
CSS + JS)



Fetch API



PHP API Layer
(auth.php / game.php /
admin.php)



PDO Prepared Statements



MySQL Database

Modules



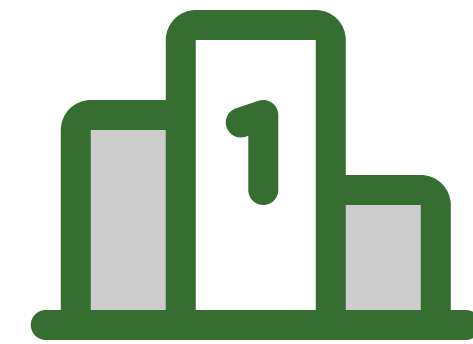
Game Module

Penalty logic and interaction



Score Management

Stores and retrieves scores



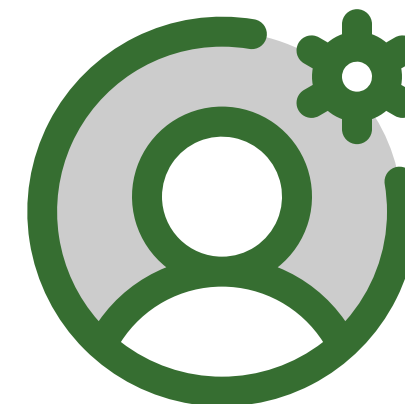
Leaderboard

Displays player rankings



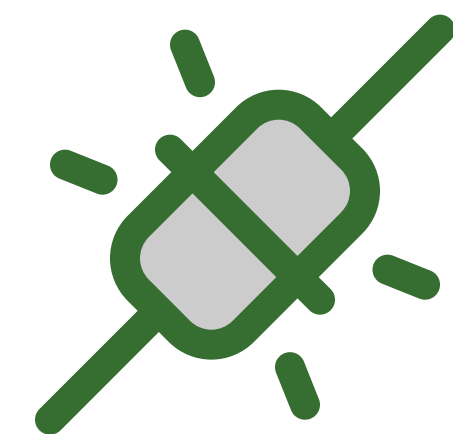
Authentication

User login and access control



Admin Panel

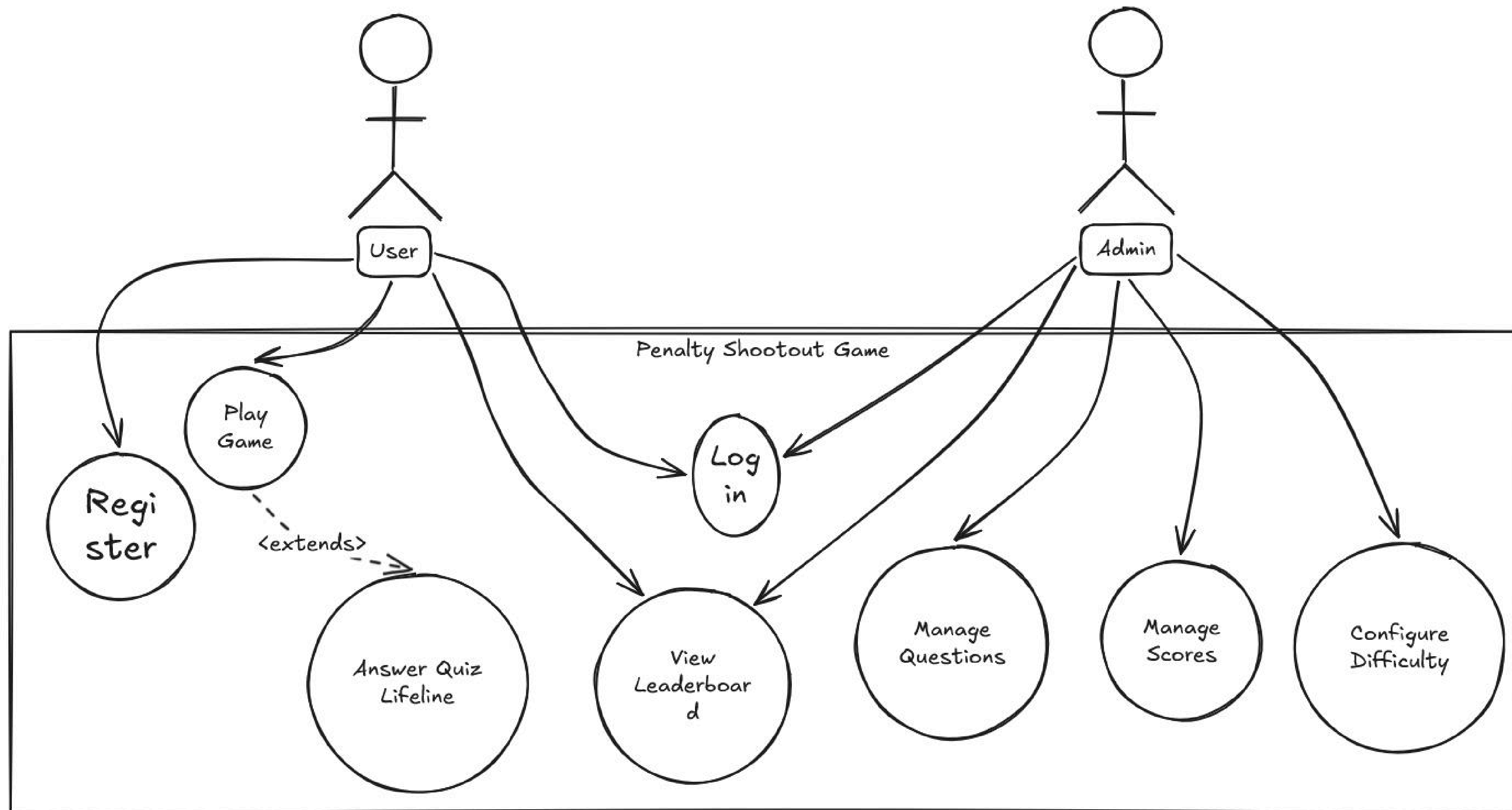
Manage questions, scores, settings



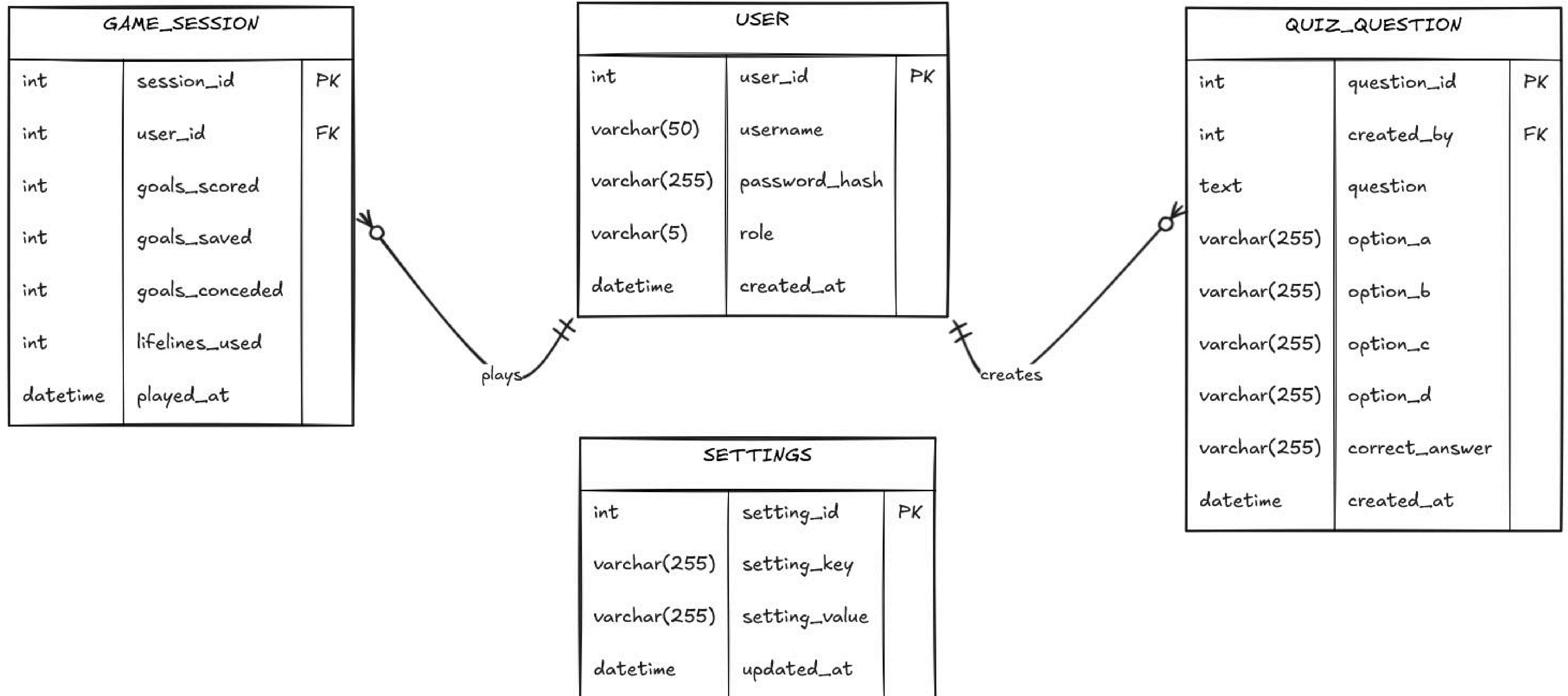
API Module

Connects frontend and backend

Use Case Diagram



ER Diagram



Working

Player is kicker:

- Player aims cursor over goal → holds Space to fill power bar → releases to shoot
- Goalkeeper AI dives — accuracy depends on difficulty set by admin
- If player misses → quiz modal appears with a football trivia question
- Correct answer → get a 2nd life, game continues
- Wrong answer → player becomes the goalkeeper and the kicker is AI



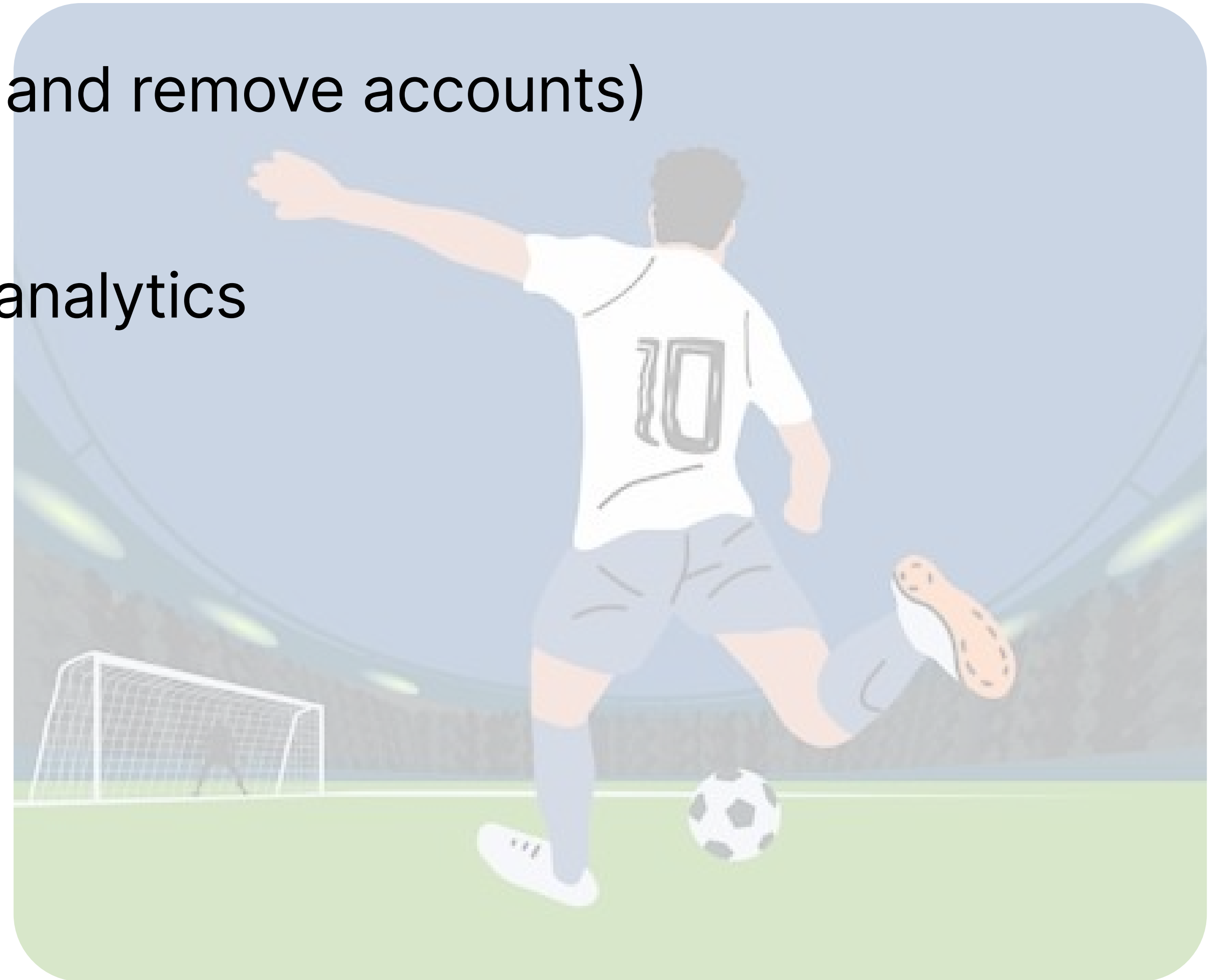
Player is goalkeeper:

- If player concedes → quiz modal appears with a football trivia question
- Correct answer → get a 2nd life, game continues
- Wrong answer → game over, session saved to database



Future Implementation

- User management in admin panel (list and remove accounts)
- Per-round data recording for detailed analytics
- Multiplayer mode using WebSockets
- Mobile-responsive game controls
- Public deployment with HTTPS



Key Takeaways

- Role-based access needs to be enforced server-side — client checks alone are not enough
- Separating frontend, API, and database makes code much easier to debug
- A quiz mechanic completely changes how a game feels
- User-friendly interface

